

Research on growth hormone, a small peptide released from the anterior pituitary gland, suggests that it affects most, if not all, cells in the body. In addition to its effects on general growth, growth hormone influences several other physiological processes.

To study the effects of growth hormone on metabolism, researchers enrolled 10 individuals (five men and five women) between the ages of 25 and 50. The individuals studied had typical (normal to slightly above average) body dimensions and compositions and did not have metabolic disease. All women were studied during the same phase of the menstrual cycle.

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In addition, mitochondria were isolated from skeletal muscle biopsies and incubated in solutions containing ADP, inorganic phosphate (P_i), reagents for the detection of ATP synthesis, and one of the following sets of substrates: α -ketoglutarate, pyruvate plus malate (malate inhibits donation of electrons to complex II of the electron transport chain), palmitoylcarnitine plus malate, glutamate plus malate, and succinate plus rotenone (rotenone inhibits complex I).

The results of the study are shown in Table 1. The treatments were compared using paired t tests.

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Free fatty acid concentration (μ M)	360 ± 74	831 ± 47	0.001
Lysine concentration (μ M)	200 ± 20	198 ± 15	0.810
Leucine concentration (μ M)	168 ± 9	158 ± 12	0.510
Leucine oxidation (μ mol/kg bodyweight/h)	42.0 ± 2.4	27.1 ± 1.0	0.002
<i>In vitro</i> results (mitochondrial ATP synthesis rate with the substrates listed below)			
α -Ketoglutarate (μ mol/min/g muscle)	4.33 ± 0.25	5.82 ± 0.53	0.049
Pyruvate plus malate (μ mol/min/g muscle)	3.24 ± 0.22	3.33 ± 0.27	0.833
Palmitoylcarnitine plus malate (μ mol/min/g muscle)	1.84 ± 0.23	3.11 ± 0.31	0.007
Glutamate plus malate (μ mol/min/g muscle)	7.06 ± 0.69	9.26 ± 0.71	0.036
Succinate plus rotenone (μ mol/min/g muscle)	2.83 ± 0.21	4.36 ± 0.42	0.014

According to the data in the passage, production of which of the following metabolites was most likely inhibited by infusion of growth hormone?

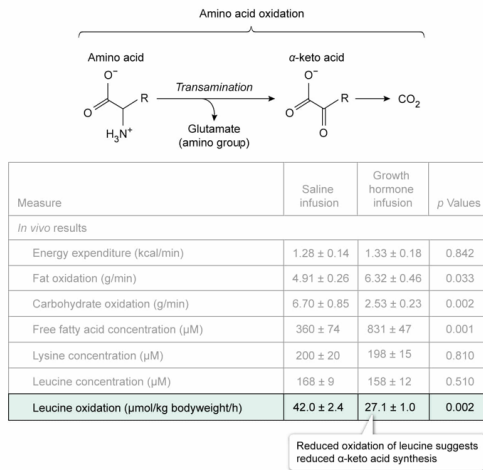
- ☐ A. α -Keto acids derived from lysine [4%]
- ✓ ☒ B. α -Keto acids derived from leucine [61%]
- ☐ C. Fatty acids derived from triglycerides [17%]
- ☐ D. Ketone bodies derived from triglyceride fatty acids [16%]

Correct

61% answered correctly

Explanation:

Reduced leucine oxidation suggests growth hormone infusion reduced α -keto acid formation



In addition to their use for synthesizing proteins, amino acids can act as signaling molecules, sources of nitrogen, or substrates that can be oxidized for energy. When amino acids are used as an energy source, their carbon skeletons are converted to glucose or ketone bodies or the amino acids are broken down and oxidized in the citric acid (Krebs) cycle.

The question asks which metabolite most likely had its production inhibited by growth hormone infusion in the study presented in the passage. The **first step** in the **utilization of amino acids for energy** is a **transamination reaction that forms the α -keto acid** (carbon skeleton) of the amino acid. **Leucine oxidation** was significantly **reduced by growth hormone** infusion, **suggesting** that production of **leucine's α -keto acid** was **inhibited** by growth hormone.

(Choice A) Lysine's concentration, like leucine's, did not change significantly in response to growth hormone infusion. However, unlike leucine, for which oxidation was measured and known to be affected by growth hormone, lysine and its oxidation are not addressed in the passage. Therefore, the effects of growth hormone infusion on the formation of lysine's α -keto acid are less certain.

(Choice C) The marked increase in free fatty acid concentration in response to growth hormone infusion suggests an *increase* in fatty acid release from hydrolyzed triglycerides.

(Choice D) The increase in free fatty acid concentration that occurred during growth hormone infusion suggests that, if anything, ketogenesis from **free fatty acids** would be *stimulated* rather than inhibited. Likewise, the reduction in leucine oxidation could theoretically enable more use of leucine's α -keto acid for ketogenesis (ie, ketone body production), but to the extent this occurred it too would *increase* rather than decrease ketogenesis.

Educational objective:

Amino acids are converted to α -keto acids (amino acid carbon skeletons) through transamination reactions.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403288

System:
Metabolic Reactions

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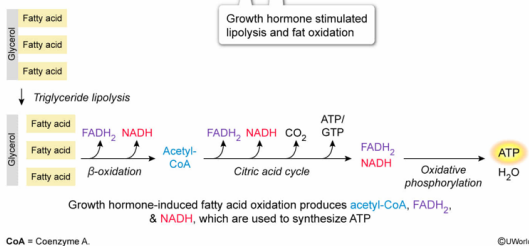
Given the *in vivo* results from the study, the effect of growth hormone on fat oxidation directly results in all of the following EXCEPT:

- ☐ A. increased acetyl-coenzyme A production from triglyceride fatty acids. [23%]
- ☒ B. increased glycerol release from triglycerides. [40%]
- ☐ C. increased NADH production from fatty acids. [11%]
- ☐ D. increased $FADH_2$ production from fatty acids. [24%]

Explanation:

The growth hormone-induced increase in fat oxidation increased production of acetyl CoA, NADH, & FADH₂ from fatty acids

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The fatty acids bound to glycerol in **triglycerides** (eg, in adipose tissue cells) represent the body's densest store of metabolic potential energy. These fatty acids can be **released from glycerol** by lipases, thereby contributing to the "free" (ie, nonesterified) fatty acids present in body fluids.

To convert the potential energy of fatty acids into the metabolic currency (eg, ATP, GTP) used by cells, free fatty acids are first **converted to fatty acyl-coenzyme A (CoA)** molecules, after which they are progressively shortened during the process of fat oxidation. Each round of fatty acid shortening **produces acetyl-CoA** and the **cofactors NADH and FADH₂**.

Table 1 shows that **growth hormone** infusion **increased** the **oxidation** of free **fatty acids**, and the question asks for the response that is *not* a direct result of this increase in fat oxidation. Although the increased concentration of free fatty acids during growth hormone infusion suggests greater release of glycerol from hydrolyzed triglycerides, **lipolysis is not part of fat oxidation**. Therefore, **glycerol release from hydrolyzed triglycerides is not a direct result of the stimulation of fat oxidation by growth hormone**.

(Choices A, C, and D) Because fat oxidation directly produces **acetyl-CoA, NADH, and FADH₂**, oxidation of fatty acids from any source directly stimulates the production of these three molecules from fatty acids.

Educational objective:

Oxidation of fatty acids produces acetyl-coenzyme A and the cofactors NADH and FADH₂.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403289
System:
Metabolic Reactions

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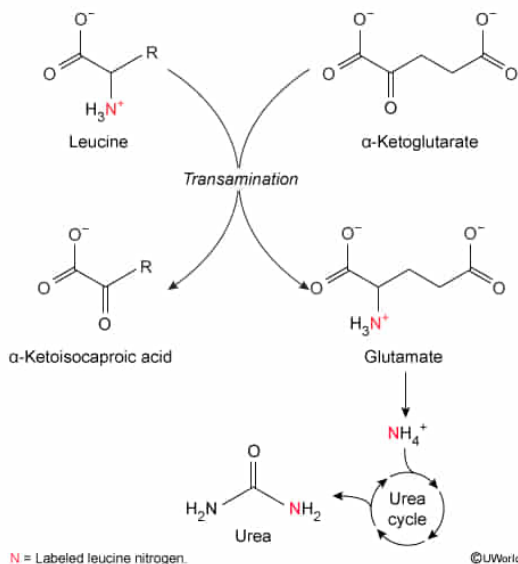
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If the leucine molecules given to participants were labeled with heavy nitrogen isotopes to monitor the contribution of leucine to urea production, the researchers would in theory have been able to detect the label:

- ☐ A. only in urea. [16%]
- ✓ ☒ B. first in glutamate and then in urea. [21%]
- ☐ C. first in the α -keto acid of leucine and then in urea. [50%]
- ☐ D. first in urea and then in citric acid cycle intermediates. [12%]

Explanation:

The label in leucine would have been detectable first in glutamate & later in urea



During the first step in oxidative degradation, amino acids undergo a **transamination reaction** that separates the amino group from the carbon skeleton. The amino group is subsequently converted to the end product urea in a series of reactions known as ureagenesis, whereas the carbon skeleton (an α -keto acid) can be directed to one of several fates, including **gluconeogenesis**, **ketogenesis**, or oxidation in the citric acid (Krebs) cycle.

The question asks where (ie, in which molecules) the nitrogen atoms labeled for the purpose of measuring urea production from leucine would have been detectable. The **amino nitrogen** in leucine **must be labeled** to facilitate this detection. This amino group is first transferred to ketoglutarate to form glutamate. The amino group is then removed from glutamate in subsequent reactions to produce urea. Therefore, the labeled atom would be detectable first in **glutamate** and then in **urea**.

(Choice A) Because the amino group of leucine is first transferred to glutamate during ureagenesis, the label can also be detected in glutamate prior to being detected in urea.

(Choices C and D) The α -keto acid of leucine is its carbon skeleton and therefore does not contain the amino group atoms that are eventually transferred to urea during ureagenesis. Although atoms from leucine can be transferred to citric acid cycle intermediates, this transfer occurs *after* the amino group is removed. Therefore, the amino group label originally in leucine is not transferred to citric acid cycle intermediates during ureagenesis.

Educational objective:

In the transamination reaction that initiates amino acid oxidation, an amino acid is deaminated to yield an α -keto acid and glutamate. The amino group transferred to glutamate in this reaction can be used to form urea in the reactions of ureagenesis.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403290

System:
Metabolic Reactions

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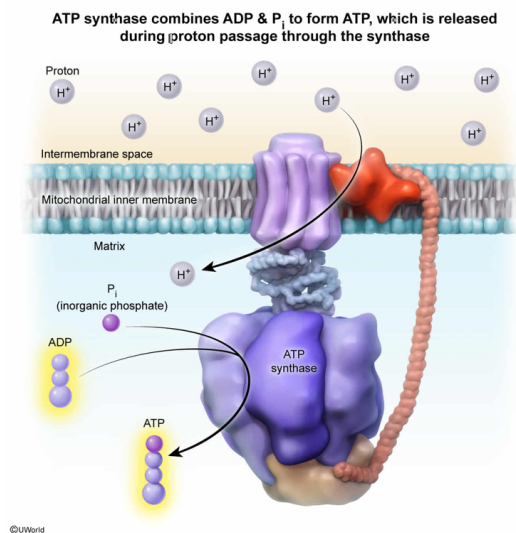
Why did the investigators include ADP in the incubation solution for the experiments measuring the effect of growth hormone infusion on mitochondrial ATP synthesis?

- ☐ A. To provide a source of energy to ATP synthase [5%]
- ✓ ☒ B. To provide a substrate for the reaction catalyzed by ATP synthase [75%]
- ☐ C. To ensure continued substrate-level phosphorylation essential for acetyl-coenzyme A production [14%]
- ☐ D. To act as an allosteric inhibitor of glycolytic ATP production [3%]

Correct

75% answered correctly

Explanation:



Mitochondrial ATP synthesis is carried out by the mitochondrial ATP synthase (complex V), which is a protein complex embedded in the mitochondrial inner membrane. The synthase contains a subunit that can exist in three states: 1) empty, 2) bound to ADP and P_i , and 3) bound to ATP. The empty subunit binds ADP and P_i and converts them to ATP in a near-equilibrium reaction. However, the newly formed ATP remains bound to ATP synthase until rotation of one part of the complex causes the ATP to be released. This rotation is caused by protons passing through ATP synthase down their concentration gradient (ie, from the intermembrane space to the matrix).

The question asks why ADP was included in the incubation solution for the experiments in which mitochondrial ATP synthesis was measured. Because **ATP is synthesized from ADP** (and P_i) bound to ATP synthase, **ADP is essential for mitochondrial ATP synthesis** to occur.

(Choice A) The energy for ATP synthesis is provided by the **proton gradient** across the inner mitochondrial membrane (*not* ADP).

(Choice C) **Substrate-level phosphorylation** occurs in the *cytosol* when **acetyl-coenzyme A** is produced through glycolysis. However, the investigators measured ATP synthesis in *isolated mitochondria* (ie, in the absence of the cytosol), so the ADP would be used for ATP synthesis by *oxidative phosphorylation*.

(Choice D) Because the measurements were conducted in isolated mitochondria, glycolysis (and glycolytic ATP production) would not occur.

Educational objective:

Mitochondrial ATP synthesis is carried out by ATP synthase as protons pass through it. This passage of protons from the mitochondrial intermembrane space to the matrix causes movement within ATP synthase that facilitates phosphorylation of ADP to form ATP.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403291

System:
Metabolic Reactions

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Based on the *in vitro* data presented in Table 1, growth hormone infusion affects mitochondrial ATP synthesis fueled by molecules derived from which of the following precursors?

- I. Proteins
- II. Carbohydrates
- III. Fats

- ☐ A. II only [10%]
- ☐ B. III only [13%]
- ☐ C. I and II only [27%]
- ✓ ☒ D. I and III only [48%]

Correct

49% answered correctly

Explanation:

***In vitro* growth hormone treatment affects substrates used for skeletal muscle mitochondrial ATP synthesis**

Measure	Saline infusion	Growth hormone infusion	p Values	Macronutrients from which substrates are generated		
<i>In vitro</i> results (mitochondrial ATP synthesis rate with the substrates listed below)				Carbohydrate	Protein	Fat
α -Ketoglutarate ($\mu\text{mol}/\text{min}/\text{g}$ muscle)	4.33 ± 0.25	5.82 ± 0.53	0.049	✓	✓	✓
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Synthesis of high-energy triphosphates (ie, ATP and GTP) by mitochondria can occur via **substrate-level phosphorylation** in the **citric acid (Krebs) cycle** (producing GTP or ATP) or via oxidative phosphorylation (producing ATP). Amino acids in proteins, fatty acids in fats, and glucose monomers from carbohydrates **all contribute** to the citric acid cycle and the production of **cofactors** that donate electrons to the electron transport chain. Hormones can influence the mixture of substrates used for mitochondrial ATP synthesis.

The passage describes a study in which individuals received infusions of either growth hormone or saline (control) to study the effects of growth hormone on certain aspects of metabolism. As part of the study, mitochondrial ATP synthesis by isolated skeletal muscle mitochondria incubated with various

substrates and inhibitors was determined.

The question asks for the fuel molecules influenced by growth hormone based on the measurements of *in vitro* mitochondrial ATP synthesis. Table 1 shows that ATP synthesis during incubation with either palmitoylcarnitine (derived from fat) or glutamate (an amino acid derived from proteins) plus malate was significantly affected by growth hormone treatment. Therefore, the ***in vitro* data suggest** that the use of molecules derived from **protein (Number I)** and **fat (Number III)** for mitochondrial **ATP synthesis** was **affected by growth hormone** infusion.

(Number II) Carbohydrates are converted to pyruvate before they are used for the citric acid cycle and ATP synthesis in mitochondria. Table 1 shows that pyruvate plus malate results in approximately the same level of ATP synthesis in the presence and absence of growth hormone. The difference between the two scenarios is statistically insignificant ($p > 0.05$). Therefore, growth hormone infusion does not affect mitochondrial metabolism of molecules derived from carbohydrates. Note that changes in the use of the citric acid cycle intermediates succinate and α -ketoglutarate for ATP synthesis could occur as part of growth hormone-induced changes in **noncarbohydrate** (ie, **fat** or **protein**) metabolism.

Educational objective:

Carbohydrates, proteins, and fats can all be metabolized to contribute to mitochondrial ATP synthesis. Many hormones influence metabolic processes such as mitochondrial ATP synthesis.

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Measure	Saline infusion	Growth hormone infusion	<i>p</i> Values
<i>In vivo</i> results			
Energy expenditure (kcal/min)	1.28 ± 0.14	1.33 ± 0.18	0.842
Fat oxidation (g/min)	4.91 ± 0.26	6.32 ± 0.46	0.033
Carbohydrate oxidation (g/min)	6.70 ± 0.85	2.53 ± 0.23	0.002
Free fatty acid concentration (μM)	360 ± 74	831 ± 47	0.001
Lysine concentration (μM)	200 ± 20	198 ± 15	0.810
Leucine concentration (μM)	168 ± 9	158 ± 12	0.510
Leucine oxidation (μmol/kg bodyweight/h)	42.0 ± 2.4	27.1 ± 1.0	0.002
<i>In vitro</i> results (mitochondrial ATP synthesis rate with the substrates listed below)			
α-Ketoglutarate (μmol/min/g muscle)	4.33 ± 0.25	5.82 ± 0.53	0.049
Pyruvate plus malate (μmol/min/g muscle)	3.24 ± 0.22	3.33 ± 0.27	0.833
Palmitoylcarnitine plus malate (μmol/min/g muscle)	1.84 ± 0.23	3.11 ± 0.31	0.007
Glutamate plus malate (μmol/min/g muscle)	7.06 ± 0.69	9.26 ± 0.71	0.036
Succinate plus rotenone (μmol/min/g muscle)	2.83 ± 0.21	4.36 ± 0.42	0.014

The results presented in Table 1 suggest that the effect of growth hormone on ATP synthesis in skeletal muscle mitochondria primarily involved which metabolic pathways?

- ☐ A. The pyruvate dehydrogenase reaction [4%]
- ☐ B. Citric acid cycle reactions that produce FADH_2 but not NADH [14%]
- ☐ C. Citric acid cycle reactions that produce NADH but not FADH_2 [29%]
- ✔ ☒ D. Entry into the citric acid cycle from nonglycolytic pathways [51%]

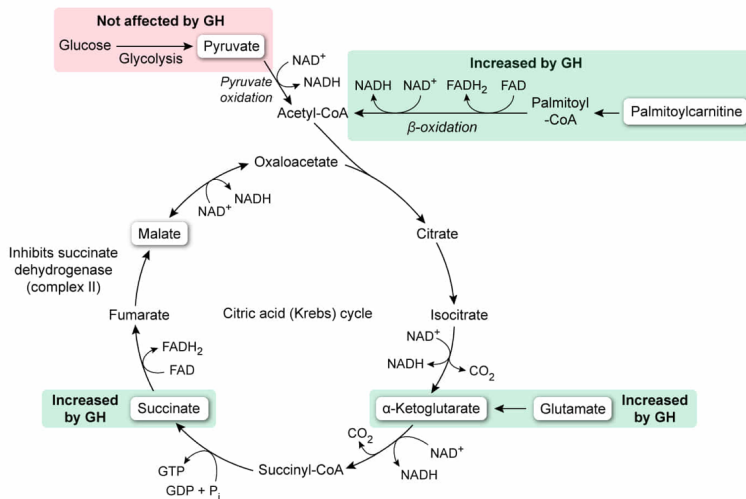
Correct

51% answered correctly

Explanation:

Growth hormone increases nonglycolytic entry into the citric acid cycle

Measure	Saline infusion	Growth hormone infusion	p Values	Carrier by which electrons enter ETC under in vitro conditions	
<i>In vitro</i> results (mitochondrial ATP synthesis rate with the substrates listed below)				NADH	FADH ₂
α -ketoglutarate ($\mu\text{mol/min/g}$ muscle)	4.33 ± 0.25	5.82 ± 0.53	0.049	✓	
Pyruvate plus malate ($\mu\text{mol/min/g}$ muscle)	3.24 ± 0.22	3.33 ± 0.27	0.833	✓	
Palmitoylcarnitine plus malate ($\mu\text{mol/min/g}$ muscle)	1.84 ± 0.23	3.11 ± 0.31	0.007	✓	✓
Glutamate plus malate ($\mu\text{mol/min/g}$ muscle)	7.06 ± 0.69	9.26 ± 0.71	0.036	✓	
Succinate plus rotenone ($\mu\text{mol/min/g}$ muscle)	2.83 ± 0.21	4.36 ± 0.42	0.014		✓



CoA = Coenzyme A; ETC = Electron transport chain; GH = Growth hormone.

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The **citric acid (Krebs) cycle** is a series of metabolic reactions that **oxidize** certain molecules to produce the reduced electron carriers **NADH** and **FADH₂** as well as the high-energy molecule **GTP**. The reduced electron carriers then contribute the electrons necessary for ATP production to the **electron transport chain**. Various **molecules can enter the citric acid cycle at different points**, and different precursor molecules must undergo different metabolic pathways to enter the cycle.

The passage presents findings from a study in which the effect of growth hormone on ATP synthesis by isolated mitochondria was determined. The question asks, based on the rates of ATP synthesis measured with various substrates, which metabolic pathways were primarily involved in the response to growth hormone. The data in Table 1 show that **ATP synthesis** increased

in the presence of growth hormone when the substrates used were not part of glycolysis. Therefore, **growth hormone likely increased entry** into the **citric acid cycle** by nonglycolytic pathways to increase the production of NADH and FADH₂, which subsequently increased ATP synthesis.

(Choice A) ATP synthesis was not significantly different in growth hormone–treated and control mitochondria when pyruvate was used as a substrate. This finding suggests that growth hormone did not affect the activity of pyruvate dehydrogenase.

(Choices B and C) Table 1 shows that growth hormone affected the ATP synthesis rate when NADH entry into complex I was inhibited by rotenone *and* when FADH₂ entry into complex II was inhibited by malate. These data suggest that levels of *both* NADH and FADH₂ produced in citric acid cycle reactions were increased in response to growth hormone treatment.

Educational objective:

Different molecules can enter the citric acid cycle at different points. The cycle oxidizes these molecules to produce the electron carriers NADH and FADH₂. Different entry points into the citric acid cycle may be inhibited by different molecules.

Time spent:0 sec

Subject:
Biochemistry

QID:403293

System:
Metabolic Reactions